Kalman filter and observability matrix, step by step (first-order gain model)

Model

$$\delta x_1[k+1] = \delta x_2[k]$$
$$\delta x_2[k+1] = \delta x_3[k] \qquad y_1[k] = \delta x_m[k] = \delta x_1[k] + n_x[k]$$
$$\delta x_3[k+1] = \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] - \varepsilon[k] \qquad y_2[k] = a_{xm}[k] = a_x[k] + n_a[k]$$
$$a_x[k+1] = a_x[k]$$

$$\mathbf{e} = \begin{bmatrix} e_{\delta x_1} \\ e_{\delta x_2} \\ e_{\delta x_3} \\ e_{a_x} \end{bmatrix}, \quad \mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} \mathbf{H}_1 \\ \mathbf{H}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{Q} = \mathrm{diag}(0, 0, Q_{33}, 0), \quad \mathbf{R} = \mathrm{diag}(R_1, R_2), \qquad Q_{33} = \mathrm{Var}(\varepsilon), \quad R_1 = \mathrm{Var}(n_x), \quad R_2 = \mathrm{Var}(n_a)$$

$$F_{dx} = \big(1 + (1 - \lambda_c)d\big)\,\frac{\Delta x_d}{\hat{a}_x} \qquad \text{(constant velocity)}$$

1 Part 1: observability matrix

1.1 What the sensors show: measurement minus its prediction

$$e_{y_1}[k] = y_1[k] - \delta\hat{x}_1[k] = \big(\delta x_1[k] - \delta\hat{x}_1[k]\big) + n_x[k] = e_{\delta x_1}[k] + n_x[k]$$
$$e_{y_2}[k] = y_2[k] - \hat{a}_x[k] \quad = \big(a_x[k] - \hat{a}_x[k]\big) + n_a[k] \quad = e_{a_x}[k] + n_a[k]$$

1.2 How an estimation error moves from one step to the next (no update, no noise)

$$e_{\delta x_1}[k+1] = e_{\delta x_2}[k]$$
$$e_{\delta x_2}[k+1] = e_{\delta x_3}[k]$$
$$e_{\delta x_3}[k+1] = \lambda_c \cdot e_{\delta x_3}[k] - F_{dx}[k] \cdot e_{a_x}[k]$$
$$e_{a_x}[k+1] = e_{a_x}[k]$$

1.3 e_{y_1} at step $k_0 + j$, written with the errors at step k_0 (1.1 then 1.2 applied j times)

$$
\begin{aligned}
e_{y_1}[k_0] &= e_{\delta x_1}[k_0] \\
&= 1 \cdot e_{\delta x_1}[k_0] + 0 \cdot e_{\delta x_2}[k_0] + 0 \cdot e_{\delta x_3}[k_0] + 0 \cdot e_{a_x}[k_0] \\
e_{y_1}[k_0+1] &= e_{\delta x_1}[k_0+1] = e_{\delta x_2}[k_0] \\
&= 0 \cdot e_{\delta x_1}[k_0] + 1 \cdot e_{\delta x_2}[k_0] + 0 \cdot e_{\delta x_3}[k_0] + 0 \cdot e_{a_x}[k_0] \\
e_{y_1}[k_0+2] &= e_{\delta x_1}[k_0+2] = e_{\delta x_2}[k_0+1] = e_{\delta x_3}[k_0] \\
&= 0 \cdot e_{\delta x_1}[k_0] + 0 \cdot e_{\delta x_2}[k_0] + 1 \cdot e_{\delta x_3}[k_0] + 0 \cdot e_{a_x}[k_0] \\
e_{y_1}[k_0+3] &= e_{\delta x_3}[k_0+1] = \lambda_c\, e_{\delta x_3}[k_0] - F_{dx}[k_0]\, e_{a_x}[k_0] \\
&= 0 \cdot e_{\delta x_1}[k_0] + 0 \cdot e_{\delta x_2}[k_0] + \lambda_c \cdot e_{\delta x_3}[k_0] - F_{dx}[k_0] \cdot e_{a_x}[k_0] \\
e_{y_1}[k_0+4] &= e_{\delta x_3}[k_0+2] = \lambda_c\, e_{\delta x_3}[k_0+1] - F_{dx}[k_0+1]\, e_{a_x}[k_0+1] \\
&= \lambda_c\big(\lambda_c\, e_{\delta x_3}[k_0] - F_{dx}[k_0]\, e_{a_x}[k_0]\big) - F_{dx}[k_0+1]\, e_{a_x}[k_0] \\
&= 0 \cdot e_{\delta x_1}[k_0] + 0 \cdot e_{\delta x_2}[k_0] + \lambda_c^2 \cdot e_{\delta x_3}[k_0] - \big(\lambda_c F_{dx}[k_0] + F_{dx}[k_0+1]\big) \cdot e_{a_x}[k_0] \\
e_{y_1}[k_0+j] &= 0 \cdot e_{\delta x_1}[k_0] + 0 \cdot e_{\delta x_2}[k_0] + \lambda_c^{j-2} \cdot e_{\delta x_3}[k_0] - \Big(\textstyle\sum_{i=0}^{j-3} \lambda_c^{j-3-i} F_{dx}[k_0+i]\Big) \cdot e_{a_x}[k_0], \qquad j \geq 3
\end{aligned}
$$

1.4 e_{y_2} at step $k_0 + j$

$$
e_{y_2}[k_0+j] = e_{a_x}[k_0+j] = e_{a_x}[k_0] = 0 \cdot e_{\delta x_1}[k_0] + 0 \cdot e_{\delta x_2}[k_0] + 0 \cdot e_{\delta x_3}[k_0] + 1 \cdot e_{a_x}[k_0]
$$

1.5 The coefficients, collected: observability matrix O

$$
\begin{bmatrix} e_{y_1}[k_0] \\ e_{y_2}[k_0] \\ e_{y_1}[k_0+1] \\ e_{y_2}[k_0+1] \\ e_{y_1}[k_0+2] \\ e_{y_2}[k_0+2] \\ e_{y_1}[k_0+3] \\ e_{y_2}[k_0+3] \\ e_{y_1}[k_0+4] \\ \vdots \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \lambda_c & -F_{dx}[k_0] \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \lambda_c^2 & -\lambda_c F_{dx}[k_0] - F_{dx}[k_0+1] \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}}_{\mathbf{O}} \begin{bmatrix} e_{\delta x_1}[k_0] \\ e_{\delta x_2}[k_0] \\ e_{\delta x_3}[k_0] \\ e_{a_x}[k_0] \end{bmatrix} + \text{noise}
$$

row j of $\mathbf{O} = \mathbf{H}\,\mathbf{\Phi}(k_0+j, k_0), \qquad \mathbf{\Phi}(k_0+j, k_0) = \mathbf{F}_e[k_0+j-1] \cdots \mathbf{F}_e[k_0]$

1.6 The e_{a_x} column alone (F_{dx} constant)

$$e_{y_2}[k_0+j] = 1\cdot e_{a_x}[k_0] + n_a \qquad (j\ge 0)$$
$$e_{y_1}[k_0+j] = c_{1,j}\cdot e_{a_x}[k_0] + (\text{position errors}) + n_x \qquad (j\ge 3)$$
$$c_{1,j} = -F_{dx}\big(1+\lambda_c+\cdots+\lambda_c^{j-3}\big) \;\to\; -\frac{F_{dx}}{1-\lambda_c}$$

2 Part 2: information on a_x

2.1 One unknown e, one reading $z = c\,e + n$, $\mathrm{Var}(n) = R$, prior variance P

$$J(e) = \frac{e^2}{P} + \frac{(z-c\,e)^2}{R}$$
$$\frac{dJ}{de} = \frac{2e}{P} - \frac{2c\,(z-c\,e)}{R} = 0 \;\Rightarrow\; \hat{e} = \frac{c\,z/R}{1/P + c^2/R}$$
$$\frac{1}{P_{\text{new}}} = \frac{1}{2}\frac{d^2J}{de^2} = \frac{1}{P} + \frac{c^2}{R}$$

2.2 y_2 readings of e_{a_x} ($c = 1$)

$$\frac{1}{P_{aa}[k]} = \frac{1}{P_{aa}[k-1]} + \frac{1^2}{R_2} \;\Rightarrow\; \frac{1}{P_{aa}[k_0+N]} = \frac{1}{P_{aa}[k_0]} + \frac{N}{R_2}$$

2.3 y_1 readings of e_{a_x} ($c = c_{1,j}$, position errors known)

$$\frac{1}{P_{aa}[k_0+N]} = \frac{1}{P_{aa}[k_0]} + \sum_{j=3}^{N-1}\frac{c_{1,j}^2}{R_1}$$

2.4 Both, and all four errors at once

$$\frac{1}{P_{aa}[k_0+N]} \le \frac{1}{P_{aa}[k_0]} + \frac{N}{R_2} + \sum_{j=3}^{N-1}\frac{c_{1,j}^2}{R_1} \qquad (\text{equality when the position errors are known})$$
$$\mathbf{P}^{-1}[k_0\,|\,k_0+N-1] = \mathbf{P}^{-1}[k_0] + \mathbf{O}^\top\mathbf{R}^{-1}\mathbf{O}, \qquad \big(\mathbf{O}^\top\mathbf{R}^{-1}\mathbf{O}\big)_{aa} = \frac{N}{R_2} + \sum_j \frac{c_{1,j}^2}{R_1} \qquad (\mathbf{Q}=0)$$

3 Part 3: Kalman filter, entry by entry

3.1 Prediction of the state

$$\delta\hat{x}_1^f[k] = \delta\hat{x}_2[k-1], \quad \delta\hat{x}_2^f[k] = \delta\hat{x}_3[k-1], \quad \delta\hat{x}_3^f[k] = \lambda_c\,\delta\hat{x}_3[k-1], \quad \hat{a}_x^f[k] = \hat{a}_x[k-1]$$

3.2 Prediction of P

$$\mathbf{P}^f = \mathbf{F}_e\,\mathbf{P}\,\mathbf{F}_e^\top + \mathbf{Q}, \qquad \mathbf{r}_1 = [0\ 1\ 0\ 0],\ \ \mathbf{r}_2 = [0\ 0\ 1\ 0],\ \ \mathbf{r}_3 = [0\ 0\ \lambda_c\ -F_{dx}],\ \ \mathbf{r}_4 = [0\ 0\ 0\ 1]$$
$$P_{ij}^f = \mathbf{r}_i\,\mathbf{P}\,\mathbf{r}_j^\top + Q_{ij}$$

$$P_{11}^f = P_{22} \qquad P_{12}^f = P_{23} \qquad P_{13}^f = \lambda_c P_{23} - F_{dx}P_{2a} \quad P_{1a}^f = P_{2a}$$
$$P_{22}^f = P_{33} \qquad P_{23}^f = \lambda_c P_{33} - F_{dx}P_{3a} \quad P_{2a}^f = P_{3a}$$
$$P_{33}^f = \lambda_c^2 P_{33} - 2\lambda_c F_{dx}P_{3a} + F_{dx}^2 P_{aa} + Q_{33} \quad P_{3a}^f = \lambda_c P_{3a} - F_{dx}P_{aa} \quad P_{aa}^f = P_{aa}$$

3.3 Update with y_1 ($\mathbf{H}_1 = [1\ 0\ 0\ 0]$)

$$S_1 = P_{11}^f + R_1, \qquad \ell_{i1} = \frac{P_{i1}^f}{S_1}\ \ (i = 1,2,3,4), \qquad \ell_{41} = \frac{P_{a1}^f}{S_1}$$
$$\hat{\mathbf{x}}' = \hat{\mathbf{x}}^f + \mathbf{L}_1\big(y_1 - \delta\hat{x}_1^f\big)$$
$$P'_{ij} = P_{ij}^f - \frac{P_{i1}^f\,P_{1j}^f}{S_1}, \qquad P'_{aa} = P_{aa}^f - \frac{\big(P_{a1}^f\big)^2}{S_1}$$

3.4 Update with y_2 ($\mathbf{H}_2 = [0\ 0\ 0\ 1]$)

$$S_2 = P'_{aa} + R_2, \qquad \ell_{i2} = \frac{P'_{ia}}{S_2}, \qquad \ell_{42} = \frac{P'_{aa}}{P'_{aa} + R_2}$$
$$\hat{\mathbf{x}} = \hat{\mathbf{x}}' + \mathbf{L}_2\big(y_2 - \hat{a}'_x\big)$$
$$P_{ij} = P'_{ij} - \frac{P'_{ia}\,P'_{aj}}{S_2}, \qquad P_{aa} = P'_{aa} - \frac{\big(P'_{aa}\big)^2}{P'_{aa} + R_2} = \frac{R_2\,P'_{aa}}{P'_{aa} + R_2}$$

(3.3 then 3.4 = joint update with $\mathbf{H}$, $\mathbf{R}$ diagonal)

3.5 P_{aa} written as $1/P_{aa}$

3.4: $$\frac{1}{P_{aa}} = \frac{P'_{aa} + R_2}{R_2\,P'_{aa}} = \frac{1}{P'_{aa}} + \frac{1}{R_2}$$

3.3: $$\frac{1}{P'_{aa}} = \frac{1}{P_{aa}^f} \cdot \frac{1}{1 - \dfrac{(P_{a1}^f)^2}{P_{aa}^f\,S_1}} \approx \frac{1}{P_{aa}^f} + \frac{(P_{a1}^f)^2}{(P_{aa}^f)^2\,S_1}$$

3.2: $$\frac{1}{P_{aa}^f} = \frac{1}{P_{aa}[k-1]}$$
$$\frac{1}{P_{aa}[k]} \approx \frac{1}{P_{aa}[k-1]} + \frac{(P_{a1}^f)^2}{(P_{aa}^f)^2\,S_1} + \frac{1}{R_2}$$

4 Part 4: the link

4.1 Prediction only, starting from $P_{1a} = P_{2a} = P_{3a} = 0$, P_{aa} constant

$$P_{3a}[k_0+1] = -F_{dx}[k_0]\,P_{aa}$$
$$P_{3a}[k_0+2] = \big(-\lambda_c F_{dx}[k_0] - F_{dx}[k_0+1]\big)\,P_{aa}$$
$$P_{2a}[k_0+j] = P_{3a}[k_0+j-1], \qquad P_{1a}[k_0+j] = P_{2a}[k_0+j-1] = P_{3a}[k_0+j-2]$$

$$P_{ia}[k_0+j] = \big(\text{coefficient of } e_{a_x}[k_0] \text{ in } e_{\delta x_i}[k_0+j],\ 1.3\big)\cdot P_{aa}$$
$$P^f_{1a}[k_0+j] = \big(\text{coefficient of } e_{a_x}[k_0] \text{ in } e_{y_1}[k_0+j],\ 1.3\big)\cdot P_{aa} = c_{1,j}\,P_{aa}$$

4.2 Put into 3.3 and 3.5

$$\ell_{41} = \frac{c_{1,j}\,P_{aa}}{S_1}, \qquad \frac{(P^f_{a1})^2}{(P^f_{aa})^2\,S_1} = \frac{c^2_{1,j}}{S_1} = \frac{c^2_{1,j}}{P^f_{11}+R_1}, \qquad \frac{1}{P_{aa}[k]} \approx \frac{1}{P_{aa}[k-1]} + \frac{c^2_{1,j}}{P^f_{11}+R_1} + \frac{1^2}{R_2}$$

(compare 2.3: $\dfrac{c^2_{1,j}}{R_1}$ per y_1 row; here $R_1 \to P^f_{11}+R_1$)

5 Part 5: the truth moves – average and speed

5.1 Weight of each step (Parts 2–4)

$$w_j = \frac{1}{P_{aa}[j]} - \frac{1}{P_{aa}[j-1]} \approx \frac{c_{1,j}^2}{P_{11}^f[j] + R_1} + \frac{1}{R_2}, \qquad W_0 = \frac{1}{P_{aa}[0]}$$

5.2 Each step's readings speak about the gain at that step

$$y_2[j] = a_x[j] + n_a[j]$$

$$J(a_x) = W_0\big(a_x - \hat{a}_x[0]\big)^2 + \sum_{j=1}^{k} w_j\big(a_x - a_x[j]\big)^2$$

$$\hat{a}_x[k] = \frac{W_0\,\hat{a}_x[0] + \sum_{j=1}^{k} w_j\,a_x[j]}{W_0 + \sum_{j=1}^{k} w_j} = \kappa_0\,\hat{a}_x[0] + \sum_{j=1}^{k} \kappa_j\,a_x[j]$$

$$\kappa_j = \frac{w_j}{W_0 + \sum_{i=1}^{k} w_i}, \qquad \kappa_0 = \frac{W_0}{W_0 + \sum_{i=1}^{k} w_i}, \qquad \kappa_0 + \sum_{j=1}^{k} \kappa_j = 1$$

5.3 Estimation error

$$e_{a_x}[k] = a_x[k] - \hat{a}_x[k] = \kappa_0\big(a_x[k] - \hat{a}_x[0]\big) + \sum_{j=1}^{k} \kappa_j\,\big(a_x[k] - a_x[j]\big)$$

5.4 The change of the truth between step j and step k

$$a_x[k] - a_x[j] = \sum_{i=j}^{k-1} \big(a_x[i+1] - a_x[i]\big) = \sum_{i=j}^{k-1} a_x'\big(h_d[i]\big)\,\Delta h_d[i], \qquad a_x' = \frac{da_x}{dh}$$

$$e_{a_x}[k] = \kappa_0\big(a_x[k] - \hat{a}_x[0]\big) + \sum_{j=1}^{k} \kappa_j \sum_{i=j}^{k-1} a_x'\big(h_d[i]\big)\,\Delta h_d[i]$$

5.5 a_x' nearly constant over the weighted steps

$$e_{a_x}[k] \approx a_x'\big(h_d[k]\big)\,\Big(h_d[k] - \bar{h}_\kappa[k]\Big), \qquad \bar{h}_\kappa[k] = \kappa_0\,h_d[0] + \sum_{j=1}^{k} \kappa_j\,h_d[j]$$

5.6 Fixed height

$$h_d[j] = h_d[k]\ \ \forall j\ \Rightarrow\ a_x[j] = a_x[k]\ \Rightarrow\ e_{a_x}[k] = \kappa_0\big(a_x[k] - \hat{a}_x[0]\big)\ \xrightarrow{k\to\infty}\ 0$$

5.7 Speed: weight per R travelled (constant velocity, $\Delta h_d = (h_0 - h_b)\,T_s/T_d$)

y_2 : $\dfrac{1}{R_2} \cdot \dfrac{T_d}{(h_0 - h_b)\,T_s} \propto T_d$ (same at every height)

y_1 : $\dfrac{c_1^2}{P_{11}^f + R_1} \cdot \dfrac{T_d}{(h_0 - h_b)\,T_s}, \quad c_1 \approx -\dfrac{1 + (1 - \lambda_c)d}{1 - \lambda_c} \dfrac{\Delta h_d}{\hat{a}_x} \Rightarrow \propto \dfrac{1}{\hat{a}_x^2\,T_d}$ (largest near the wall)

κ over h : y_2 only $\Rightarrow$ shape independent of T_d; y_1 added $\Rightarrow$ extra weight near the wall $\propto 1/T_d$

Figure for 5.2–5.3: filter vs weighted average with $w_j = \Delta(1/P_{aa})$

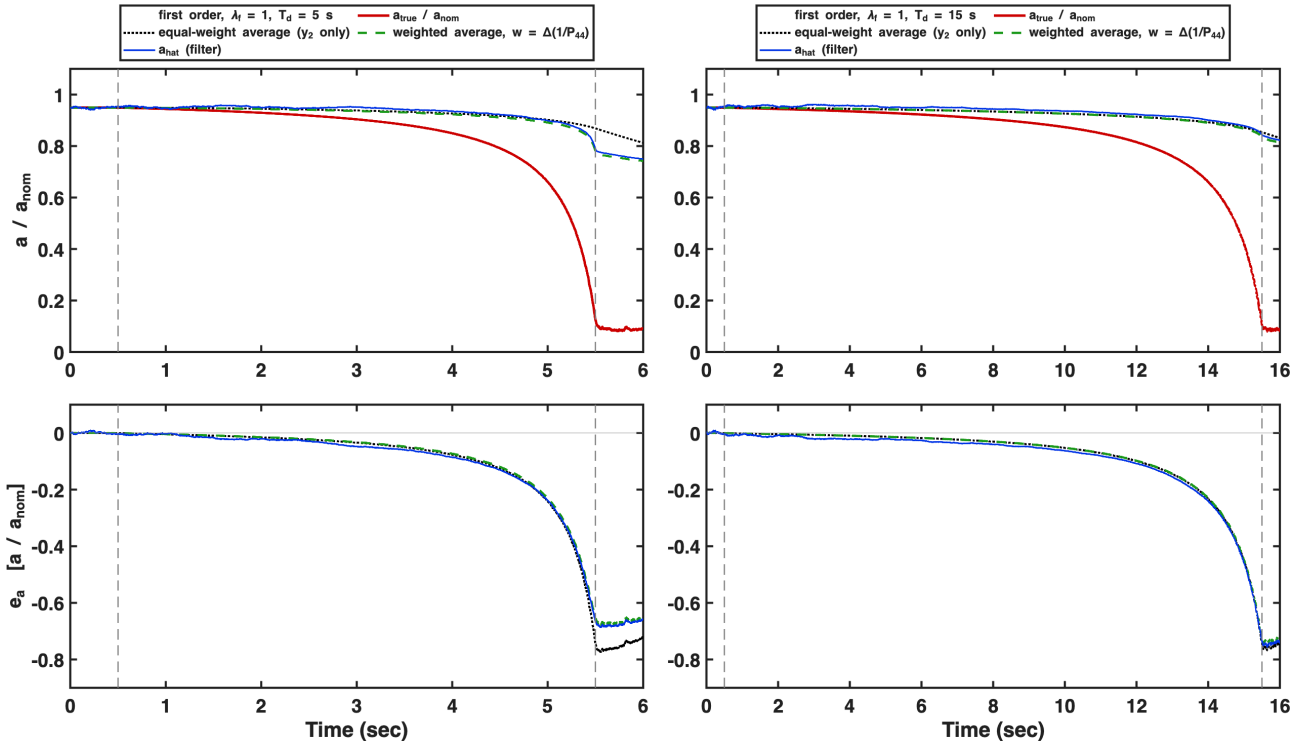


Figure for 5.7: weight added per step and per R travelled

